

# Dajin Lee, Ph. D. Candidate

Curriculum Vitae

Last Updated on April 16, 2026

## Current Position and Contact Information

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| Position   | Ph. D. Candidate<br>Department of Convergence IT Engineering<br>Pohang University of Science and Technology (POSTECH) |
| Phone      | +82-10-3383-4467                                                                                                      |
| E-mail     | <a href="mailto:dj1122@postech.ac.kr">dj1122@postech.ac.kr</a>                                                        |
| Web        | <a href="https://leerubi.github.io/">https://leerubi.github.io/</a>                                                   |
| Laboratory | <a href="https://interaction.postech.ac.kr/">https://interaction.postech.ac.kr/</a>                                   |



## Research Interests

### Haptic Perception & Rendering

- Spatial Tactile Rendering for Multiple Targets (CHI'26a)
- Discomfort from Steering Wheel Vibration (HS'26)
- Effects of Contact Force Across Upper Body (TOH'24a)
- Heart Activity Modulation Using Haptic Biofeedback (UIST'24)
- Body-Penetrating Phantom Sensation on Feet (WHC'21)

### Multisensory Integration of Sound and Touch

- Perceptual Effects of Co-Located Vibrotactile and Auditory Cues (Submitted)
- Spatial Audio-Tactile Alignment for Directional Cueing (TVCG'25)
- Modulating Body-Penetrating Tactile Sensation Using Sound (WHC'25)

### Human-Computer Interaction and Qualitative Analysis

- A Contextual Analysis of Social Touch (CHI'26b)
- Embodied and Tangible Learning for Children (IJHCS'25)
- Haptic Exhibit for Learning Physics (TOH'24b)
- Participatory Design for Vibrotactile Warning (Eurohaptics'24)

## Education

|                   |                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021.03 - Present | <b>Ph. D. Candidate, Interaction Laboratory</b><br>Computer Science and Engineering<br>Pohang University of Science and Technology (POSTECH)<br>Advisor: <a href="#">Prof. Seungmoon Choi</a> |
| 2013.03 - 2020.02 | <b>Bachelor of Science</b><br>Industrial Management and Engineering<br>Pohang University of Science and Technology (POSTECH)                                                                  |

## Publications

### International Journal Articles

- [J1] **Dajin Lee**, Gyeore Yun, and Seungmoon Choi, "Effects of Contact Force on Vibrotactile Perceived Intensity Across the Upper Body," *IEEE Transactions on Haptics*, 2024 (Also presented in the IEEE Haptics Symposium 2024; **Candidate for the Best Paper Award**).  
[Paper](#), [Presentation Slides](#)
- [J2] **Dajin Lee**, Daehyeon Nam, Jinhyuk Yoon, Dukchan Yoon, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi, "A Comparative Study of Physical and Haptic Exhibits in an Informal Learning Environment," *IEEE Transactions on Haptics*, 2024 (Also presented in the 2025 *IEEE World Haptics Conference*).  
[Paper](#), [Demo Video](#)
- [J3] **Dajin Lee** and Seungmoon Choi, "Perceptual Alignment of Spatial Auditory and Tactile Stimuli for Effective Directional Cueing," *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025 (Also presented in the 2025 *IEEE Conference on Virtual Reality and 3D User Interfaces*).  
[Paper](#), [Presentation Video](#)
- [J4] **Dajin Lee**, Daehyeon Nam, and Seungmoon Choi, "Augmenting Outdated Museum Exhibits with Embodied and Tangible Interactions for Prolonged Use and Learning Enhancement," *International Journal of Human-Computer Studies (IJHCS)*, 2025.  
[Paper](#), [Demo Video](#)
- [J5] **Dajin Lee**, Uison Ju, and Seungmoon Choi, "Perceptual Effects of Co-Located Vibrotactile and Auditory Cues on the Head," *Springer VR*, 2026 (submitted).

### International Conference Papers

- [C1] **Dajin Lee**, Seungjae Oh, Seungmoon Choi, and Bum-Jae You, "Vibrotactile Metaphor of Physical Interaction Using Body-Penetrating Phantom Sensations: Stepping on a Virtual Object," *IEEE World Haptics Conference (WHC)*, 2021.  
[Paper](#), [Presentation Video](#)
- [C2] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi, "Participatory Design for In-Vehicle Vibrotactile Warnings on Driver's Seat," *EuroHaptics*, 2024.  
[Paper](#), [Presentation Video](#), [Demo Video](#)
- [C3] Andreia Valente, **Dajin Lee**, Seungmoon Choi, Mark Billingham, and Augusto Esteves, "Modulating Heart Activity and Task Performance Using Haptic Heartbeat Feedback: A Study Across Four Body Placements," *ACM Symposium on User Interface Software and Technology (UIST)*, 2024.  
[Paper](#), [Presentation Video](#)
- [C4] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi, "Enhancing Body-Penetrating Phantom Sensations Through Multisensory Integration of Sound and Vibration," *IEEE World Haptics Conference (WHC)*, 2025.  
[Paper](#), [Supplementary Video](#)

- [C5] **Dajin Lee**, Soyeon Nam, Minseong Noh, and Seungmoon Choi, "Effects of Frequency and Magnitude on Steering Wheel Vibration Discomfort: A Hierarchical Modeling Approach to Individual Sensitivity," *IEEE Haptics Symposium*, 2026.  
[Paper](#)
- [C6] Ayush Bhardwaj, Ashish Pratap, Abbas Khawaja, Yapeng Tian, Uison Ju, **Dajin Lee**, Seungmoon Choi, and Jin Ryong Kim, "Touch with Meaning: A Contextual Analysis of Social Touch," *ACM CHI conference on Human Factors in Computing Systems (CHI)*, 2026.  
[Paper](#)

## Domestic Conference Papers (Republic of Korea)

- [DC1] Dong Yeong Jeong, Sung H. Han, Seungmoon Choi, Mingyu Lee, Haewoo Kang, Sangyoon Han, Gyeore Yun, Hyo Seung Lee, Jiwan Lee, and **Dajin Lee**, "Motion and Haptic Effects Classification of a 4D Cinema Seat," *Ergonomics Society of Korea*, 2019.
- [DC2] Daehyeon Nam, **Dajin Lee**, and Seungmoon Choi, "Improving the Design of Science Museum Exhibits Through Digital Augmentation," *Korea Software Congress (KSC)*, 2022.
- [DC3] **Dajin Lee**, Gyeore Yun, and Seungmoon Choi, "Effects of Contact Force on Vibrotactile Perceived Intensity Across Upper Body," *Korea Haptics Conference (KHC)*, 2023.
- [DC4] **Dajin Lee** and Seungmoon Choi, "Differential Thresholds of Azimuth between Auditory and Tactile Stimuli," *Korea Haptics Conference (KHC)*, 2024 (**Best Paper Award**).

## Posters & Demonstrations

- [P1] **Dajin Lee**, Daehyeon Nam, Seokwon Jeong, Keehoon Kim, and Seungmoon Choi, "A Comparative Study of Physical and Haptic Exhibits in Informal Learning Environment," *IEEE World Haptics Conference (WHC)*, 2023 (Work in Progress).
- [P2] **Dajin Lee** and Seungmoon Choi, "Spatial Tactile Optimizer: Toward Perceptual Optimization for Multi-Point Vibrotactile Cues on the Torso," *ACM CHI conference on Human Factors in Computing Systems (CHI)*, 2026 (Extended Abstract).  
[Paper](#)
- [D1] **Dajin Lee**, Jaejun Park, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi, "Participatory Design for In-Vehicle Vibrotactile Warnings on Driver's Seat," *EuroHaptics*, 2024.  
[Paper](#), [Presentation Video](#), [Demo Video](#)
- [D2] Hyunuk Kim, **Dajin Lee**, and Seungmoon Choi, "Body-Penetrating Tactile Phantom Sensations Enhanced by Sound," *IEEE World Haptics Conference (WHC)*, 2025.  
[Paper](#), [Supplementary Video](#)

## Invited Talks and Tutorials

1. Seungmoon Choi and **Dajin Lee**, "Incorporating Qualitative Analysis in Haptics Research," *IEEE World Haptics Conference (WHC)*, 2025.  
[Page](#), [Presentation Slides](#)

## Honors & Awards

- 2026 **Research Grant for Korea-Canada Research Training**  
National Research Foundation of Korea (NRF)
- 2026 **Digitally Augmented Exhibits for Local Science Museums**  
**POSTECH Metaverse Contents Contest (First Prize)**  
Pohang University of Science and Technology (POSTECH)
- 2025 **Braile Ring: An Intelligent Tactile Assistant for the Visually Impaired**  
**AWS Prize (Honorable Mention)**  
Institute of Information & communications Technology Planning & Evaluation (IITP)
- 2024 **Differential Thresholds of Azimuth between Auditory and Tactile Stimuli**  
**Best Paper Award**  
Korea Haptics Conference (KHC)
- 2024 **Effects of Contact Force on Vibrotactile Perceived Intensity Across the Upper Body**  
**Best Paper Award (Finalist)**  
IEEE Haptics Symposium
- 2023 **HapCinema: Multimedia Haptic Rendering Using In-Vehicle Haptic Interfaces**  
**Hyundai Student Innovation Challenge (Honorable Mention)**  
Korea Haptics Conference (KHC)
- 2019 **Homepage for Stary Cats at POSTECH**  
**POSTECH Hackathon (Popularity Award)**  
Pohang University of Science and Technology (POSTECH)
- 2018 **StairX: Automatic Stair Recognition and Map Service Using User's Walking Data**  
**POSTECH Hackathon (Excellence Award)**  
Pohang University of Science and Technology (POSTECH)
- 2018 **Prediction of Swell Wave Events Using Deep Learning Analysis of Time Series Data**  
**Student Innovation Challenge (Excellence Award)**  
Korean Institute of Industrial Engineers (KIIE)

## Research Experiences

### Research Internships

|                                        |                                                                                                                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019.01 - 2019.06<br>2020.03 - 2020.08 | <b>Motion and Haptic Effects Classification of a 4D Cinema Seat</b><br>Interaction Laboratory,<br>Pohang University of Science and Technology (POSTECH)<br>Advisor: Prof. Seungmoon Choi                |
| 2019.06 - 2019.08                      | <b>3D Graphical User Interface in Virtual Reality</b><br>Center for Intelligent & Interactive Robotics<br>Korea Institute of Science and Technology (KIST)<br>Advisor: Dr. Jung-Min Park                |
| 2020.09 - 2020.12                      | <b>EMG-based Prosthetic Hand Control</b><br>Medical Assistant Robotics and Cognitive Haptics (MARCH) Laboratory,<br>Pohang University of Science and Technology (POSTECH)<br>Advisor: Prof. Keehoon Kim |
| 2025.07 - 2025.08                      | <b>Development of a Taxonomy of Social Touch</b><br>Multimodal Interaction (MI) Laboratory, University of Texas at Dallas<br>Advisor: Prof. Jin Ryong Kim                                               |
| 2026.06 - 2026.12                      | <b>Perceptual Space of Gaming Sounds and Haptic Feedback</b><br>Haptic Experience Laboratory, University of Waterloo<br>Advisor: Prof. Oliver Schneider                                                 |

### Fund Projects

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019.01 - 2019.06 | <b>Automatic Authoring of Physical and Perceptual/Affective Motion Effects for Virtual Reality</b><br>Samsung Research Funding & Incubation Center for Future Technology                                                                                |
| 2021.01 - 2022.12 | <b>Development and Validation of Real-Virtual Blended Exhibits for In-Depth Embodied Learning of Scientific Principles (Team Leader)</b><br>Support Program for Reinforcement of Science Culture Exhibition Service, National Research Foundation (NRF) |
| 2022.07 - 2023.06 | <b>Research on Function Advancement to Improve the Marketability of Vibration Seat (Team Leader)</b><br>Hyundai Automobile Company                                                                                                                      |
| 2022.09 - 2026.02 | <b>Semantic Sound-to-Haptic Automatic Conversion: Metaverse, Full-body Haptic Effects, and Accessibility</b><br>Mid-Career Researcher Program, National Research Foundation (NRF)                                                                       |
| 2025.01 - Present | <b>Quantifying VOCs through Affective Natural Language Analysis of Steering Wheel Vibrations (Team Leader)</b><br>Hyundai Automobile Company                                                                                                            |

## Teaching & Mentoring Experiences

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|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2019.09 - 2021.09 | <b>Introduction to Statistics for Data Science</b> (Teaching Assistant)<br>POSTECH & POSCO Group University                         |
| 2024.09 – 2024.12 | <b>Introductions to Haptics</b> (Teaching Assistant)<br>Pohang University of Science and Technology (POSTECH)                       |
| 2025.10           | <b>Human-Computer Interaction</b> (Guest Lecturer)<br>Pohang University of Science and Technology (POSTECH)                         |
| 2025.01 – 2025.12 | <b>Undergraduate Research Program</b> (Mentor, <b>Best Project Award</b> )<br>Pohang University of Science and Technology (POSTECH) |

## Academic Service

|          |                                                 |
|----------|-------------------------------------------------|
| Reviewer | <b>Eurohaptics</b> (2024)                       |
| Reviewer | <b>IEEE Transactions on Haptics</b> (2025-2026) |
| Reviewer | <b>IEEE Haptics Symposium</b> (2026)            |
| Reviewer | <b>ACM SIGCHI</b> (2026)                        |
| Reviewer | <b>Frontiers in Psychology</b> (2026)           |

## Skills

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|--------------------------------|-----------------------------------------------------------------|
| <b>Haptic Interaction</b>      | Tactile rendering/perception, Embodied learning, Authoring tool |
| <b>Multisensory Perception</b> | Spatial audio-tactile perception, Visuo-haptic illusion         |
| <b>Programming</b>             | C/C++/C#, MATLAB, Python, R                                     |
| <b>Statistics</b>              | Statistical analysis, Design of experiment                      |